

Example of Conjunction

Find the conjunction between

“The sun is shining and it is raining.”

P = the sun is shining

Q = it is raining

P	Q	$P \wedge Q$
The sun is shining T	It is raining T	The sun is shining and it is raining. T
The sun is shining T	It is not raining F	The sun is shining and it is not raining. F
The sun is not shining F	It is raining T	The sun is not shining and it is raining. F
The sun is not shining F	It is not raining F	The sun is not shining and it is not raining. F

Example No 02;

Find the conjunction between

“I will have salad for lunch and I will have steak for dinner”

p = “I will have salad for lunch”

q = “I will have steak for dinner”

P	Q	$P \wedge Q$
I will have salad for lunch T	I will have steak for dinner T	
I will have salad for lunch T	I will not have steak for dinner F	
I will not have salad for lunch F	I will have steak for dinner T	
I will not have salad for lunch F	I will not have steak for dinner F	

P	Q	$P \wedge Q$
I will have salad for lunch T	I will have steak for dinner T	I will have salad for lunch and I will have steak for dinner T
I will have salad for lunch T	I will not have steak for dinner F	I will have salad for lunch and/but I will not have steak for dinner F
I will not have salad for lunch F	I will have steak for dinner T	I will not have salad for lunch and/but I will have steak for dinner F
I will not have salad for lunch F	I will not have steak for dinner F	I will not have salad for lunch and/but I will not have steak for dinner F

Example of Disjunction

Find the conjunction between

“Either my car has a bad engine or my car has a bad carburetor”

p = “My car has a bad engine”

q = “My car has a bad carburetor”

P	Q	$P \vee Q$
My car has a bad engine T	My car has a bad carburetor T	My car has a bad engine or a bad carburetor. T
My car has a bad engine T	My car has not bad carburetor F	car has a bad engine OR not a bad carburetor T
My car has not a bad engine F	My car has a bad carburetor T	My car has not a bad engine OR has a bad carburetor T
My car has not a bad engine F	My car has not bad carburetor F	My car has not bad engine OR bad carburetor F

Example of implication

Find the Implication between

“If I am elected in election, then I will lower taxes.”

P: If I am elected in election

Q: I will lower taxes

Case 1: if P and Q both are true, then $P \rightarrow Q$ is also true.

P	Q	$P \rightarrow Q$
If I am elected in election	I will lower the taxes	If I am elected in election, then I will lower taxes.”
T	T	T
If the politician is elected, voters would expect this politician to lower taxes.		

Case 2: if P is true and Q is false, then $P \rightarrow Q$ is False.

P	Q	$P \rightarrow Q$
If I am elected in election	I will not lower the taxes	If I am elected in election, then I will not lower taxes.”
T	F	F
It is only when the politician is elected but does not lower taxes that voters can say that the politician has broken the campaign rule.		

Case 3: if P is false and Q is true, then $P \rightarrow Q$ is True.

P	Q	$P \rightarrow Q$
If I am not elected in election	I will lower the taxes	If I am not elected in election, then I will lower taxes.”
F	T	T
If the politician is not elected, although the person may have sufficient influence to cause those in power to lower taxes.		

Case 1: if P and Q both are true, then $P \rightarrow Q$ is also true.

P	Q	$P \rightarrow Q$
If I am not elected in election	I will not lower the taxes	If I am not elected in election, then I will not lower taxes.”
F	F	T
if the politician is not elected, then voters will not have any expectation that this person will lower taxes		